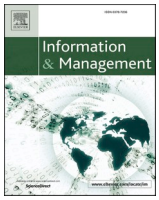
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Heterogeneous cross-side network effects on peer-to-peer platforms: Implications of nonprofessional suppliers and professional suppliers

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ABSTRACT

Cross-side network effects (CNEs) have been widely acknowledged in platform literature for their central role in shaping platform growth and governance. While growing literature focuses on the relative strengths of CNEs driven by suppliers and consumers, most studies assume that suppliers are homogeneous and thereby overlook the potential heterogeneity of CNEs driven by different suppliers. This study contributes by further distinguishing between nonprofessional and professional suppliers on P2P platforms. Using panel-vector-auto-regression (PVAR) analysis, we examine whether the strengths of CNEs between suppliers and consumers are heterogeneous across these two types of suppliers. The empirical results suggest that regarding supplier-to-consumer CNEs, both the quantity and quality of nonprofessional suppliers (vs. professional suppliers) have stronger CNEs on consumer size. In turn, regarding consumer-to-supplier CNEs, consumer size has stronger CNEs on the quantity of nonprofessional suppliers (vs. professional suppliers) but has weaker CNEs on the quality of nonprofessional suppliers (vs. professional suppliers). These findings contribute to the theoretical understanding of the dynamic nature of CNEs and practical strategies for platform governance in coordinating with nonprofessional and professional suppliers.

1. Introduction

Peer-to-peer (P2P) platforms are rapidly growing business models that function as digital intermediaries, enabling peer individuals to directly exchange goods, services, or resources with consumers [1-3]. These platforms have facilitated economic activities across various industries such as rental services (e.g., Airbnb, RVShare, Turo), freelance work (e.g., Upwork, Fiverr), caregiving (e.g., Rover, Care.com), goods exchange (e.g., Etsy, eBay, Taobao), and local services (e.g., TaskRabbit). For example, Airbnb has attracted more than 5 million hosts and more than 1.5 billion renters across the globe as of 2023 [4]. Similar to general two-sided platforms, P2P platforms are characterized by cross-side network effects (CNEs)—the phenomenon that the value derived by users on one side of the platform depends on the varying characteristics (e.g., quantity and quality) of users on the other side [5–7]. In the case of goods exchange markets like Etsy, positive seller-to-buyer CNEs occur when an increase in the quantity of sellers leads to an increase in buyer size.

CNEs have strategic implications for platform governance for both suppliers and consumers [,8–11]. First, CNEs relate to how one side drives the growth of the other side, thereby influencing a platform's strategies to build the installed base on either side. For example, a platform may determine which side receives a discount to attract the other side based on the relative strengths of CNEs on both sides [6,12]. Second, CNEs are recognized as a critical factor shaping platform competition [13]. A platform that starts with a small lead in attracting either side of the market is likely to win the entire market [14,15]. Platforms could also leverage CNEs to reinforce their market position and create significant entry barriers for new entrants [16,17].

The importance of CNEs has prompted growing literature to focus on the relative strengths of CNEs driven by suppliers and consumers (e.g., [18,19]). For example, Chu and Manchanda [20] find that supplier-to-consumer CNEs are stronger than consumer-to-supplier CNEs in C2C platforms. While these findings provide valuable insights into CNEs on platforms, they operate primarily under the assumption that all suppliers are homogeneous and thereby overlook the potential heterogeneity of CNEs driven by different suppliers. Such a nuanced gap becomes increasingly significant in light of recent practical and academic concerns about two distinct types of suppliers on P2P platforms: nonprofessional and professional suppliers.

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In practice, by lowering entry barriers and democratizing market participation, P2P platforms have inadvertently evolved into heterogeneous marketplaces that attract both nonprofessional and professional suppliers [21,22]. Managers on these platforms have increasingly recognized such supplier heterogeneity and attempted to govern it using different strategies. For example, Etsy encourages both amateur and professional sellers to sell unique products made or handpicked by themselves. Additionally, Etsy provides "Etsy Wholesale" and "Etsy Plus" for professional sellers to expand their business. Fiverr has committed different resources to attract different types of freelancers: It provides a "Fiverr Pro" plan to ensure professional freelancers' quality and a "Fiverr Learn" plan to help amateurs improve skills and service quality. Airbnb implemented a "One host, one home" policy in April 2016, which primarily regulated professional hosts' listing behaviors and stimulated the participation of nonprofessional hosts [1]. In our empirical context, the platform of real estate rental properties grants homeowners and institutional hosts (e.g., housing rental agents, real estate agents) distinct access accounts to list properties. In essence, these examples imply the potential importance of understanding and distinguishing nonprofessional and professional suppliers in the governance of P2P platforms.

Theoretically, researchers have investigated the behaviors and impacts of these two types of suppliers on P2P platforms [23-26]. Most studies find that such suppliers are systematically different in behaviors and imply the superiority of professional suppliers in platform quality and performance. For example, professional suppliers are likely to own richer resources (e.g., particular skills, money, time) needed for business than nonprofessional suppliers do [1]. Therefore, we consider that these two types of suppliers may have distinct abilities to create and capture value, thereby resulting in heterogeneous CNEs. Indeed, if the CNEs driven by different suppliers are heterogeneous, platforms could better identify which group of suppliers plays a more important role in growth and thereby improve governance strategies in terms of supplier management. For example, this would help platforms decide which group to prioritize in resource commitment and which group receives a discount to better stimulate CNEs. In sum, recognizing the potential heterogeneity of CNEs driven by different suppliers is theoretically and practically crucial, which not only deepens insights into the nature of CNEs but also informs better supplier management strategies for P2P platforms.

Motivated by the critical research gap and echoing McIntyre and Srinivasan [11]'s call for the exploration of supplier heterogeneity in CNEs, this study examines the heterogeneous CNEs driven by nonprofessional and professional suppliers. Developed from extant literature, we acknowledge that the two types of suppliers differ in three respects: identity formality, resource richness, and entry motivation. In this way, nonprofessional suppliers refer to those with informal identities and limited resources, and are motivated by the minimum supply costs. Professional suppliers refer to profit-seekers with formal identities and rich resources (e.g., time, money, specialist skills). Then we pin down four-way CNEs: the CNEs that nonprofessional suppliers have on consumers (i.e., *nonprofessional-to-consumer CNEs*), the CNEs that professional suppliers have on consumers (i.e., *professional-to-consumer CNEs*), the CNEs that consumers have on nonprofessional suppliers (i.e., *consumer-to-nonprofessional CNEs*), and the CNEs that consumers have on professional suppliers (i.e., *consumer-to-professional CNEs*). Utilizing two important characteristics on the supply side (i.e., supply quantity and quality), we consider how the quantity and quality of different suppliers impact platform attractiveness to the demand side and how consumers impact platform attractiveness to different suppliers.

Following extant literature, we identify CNEs by examining how much one side's attribute changes in response to a change in the other side's attribute [18,19]. Using a unique dataset collected from a platform in real estate properties, we conduct the panel-vector-auto-regression (PVAR) analysis with several robustness tests. Our findings reveal heterogeneous CNEs between different suppliers and consumers. Specifically, regarding the supplier-to-consumer CNEs, both the quantity and quality of nonprofessional suppliers (vs. professional suppliers) have stronger CNEs on consumer size. Regarding the consumer-to-supplier CNEs, the consumer size has stronger CNEs on the quantity of nonprofessional suppliers (vs. professional suppliers) but has weaker CNEs on the quality of nonprofessional suppliers (vs. professional suppliers). We also find suggestive results that complement such heterogeneous CNEs. On the one hand, adding nonprofessional suppliers (vs. professional suppliers) could help increase supply diversity and decrease product prices on platforms, thereby resulting in a larger attractiveness to consumers. On the other hand, adding consumers would increase supply diversity and provide suppliers with diverse opportunities, thereby resulting in a larger attractiveness to nonprofessional suppliers (vs. professional suppliers).

These findings contribute to the extant literature in several important ways. First, this study contributes to the CNEs literature in two-sided platforms by moving away from the assumption that all suppliers are homogeneous. The explorations of the heterogeneous CNEs driven by different suppliers not only deepen the nature of CNEs but also enrich platform governance strategies on how to better stimulate CNEs. Second, extant literature on nonprofessional and professional suppliers primarily investigates their impacts on platform performance and suggests the superiority of professional suppliers. We contribute by exploring their relative impacts on platform growth and revealing the superiority of nonprofessional suppliers, which provides insights into the theoretical and practical implications of different suppliers. Third, this study expands the theoretical understanding of two critical supply-side attributes—quantity and quality—in capturing CNEs. By integrating supplier types and supply-side attributes, this study offers a more nuanced perspective on how heterogeneous CNEs emerge and provides valuable insights for tailoring governance strategies to effectively manage different suppliers.

2. Literature review

2.1. CNEs in two-sided platforms

Following Webster and Watson [27], we use a concept-centric approach to review two related streams of literature. Our work first relates to the literature on CNEs in two-sided platforms. Much of this literature investigates the existence and impacts of CNEs on platform competition [8,13,10,28,29], platform growth [20,30-32], and consumer behavior [33,34]. These studies primarily highlight the critical roles of CNEs in platforms. To better leverage CNEs, researchers also focus on platform strategies such as optimal pricing [35,6,17], adding a large number of suppliers [16,36-38], and improving platform quality [39,40,15].

Some recent studies have begun to explore the nature of heterogeneous CNEs, which are directly related to our work. Most of them investigate heterogeneous CNEs by distinguishing different types of users, e.g., suppliers vs. consumers [20,19,41], and new users vs. existing users [10]. For example, in the television markets, Wilbur [41] empirically finds that in television networks, viewers (vs. advertisers) could stimulate stronger CNEs. Chu and Manchanda [20] find that suppliers could drive stronger CNEs than consumers do. Song et al. [19] further incorporate a temporal dimension and demonstrate that while suppliers (vs. consumers) mainly drive stronger CNEs in the short term, they drive weaker CNEs than consumers do in the long term. Hinz et al. [10] propose an influx-outflow model to estimate network effects, which suggests that existing and new users (either suppliers or consumers) could drive different levels of CNEs. Yuan et al. [12] model the platform's optimal pricing in the presence that supplier-to-consumer CNEs and consumer-to-supplier CNEs may be asymmetric in strength. In addition to distinguishing different types of users, some researchers investigate heterogeneous CNEs by distinguishing different types of platforms. For example, Cong et al. [18] empirically show that the

asymmetric CNEs between lenders and borrowers would vary across growing and declining platforms. By distinguishing first-mover and second-mover firms, Saito and Matsubayashi [42] model how the asymmetric network effects of such two firms would impact their product positioning strategies.

We examine heterogeneous CNEs but differ from these studies. Specifically, most literature on heterogeneous CNEs primarily distinguishes suppliers and consumers, assuming all suppliers are homogeneous. This study contributes by moving away from such an assumption and further distinguishing two types of suppliers (i.e., nonprofessional and professional suppliers). Our empirical findings suggest that the strengths of both supplier-to-consumer CNEs and consumer-to-supplier CNEs would vary across nonprofessional and professional suppliers. Such a nuanced perspective deepens the theoretical understanding of CNEs by providing more precise insights into heterogeneous CNEs among different users.

2.2. Nonprofessional and professional suppliers

Our work also relates to the literature on nonprofessional and professional suppliers on P2P platforms. Given the prevalent phenomenon that nonprofessional and professional suppliers coexist on the supply side, most literature primarily focuses on the behaviors of such distinct suppliers. For example, some early studies began by investigating their behaviors separately [43,44]. Recently, a growing number of studies have examined their behaviors by comparing their differences in entry motivation [45,46], pricing [47,48,23,25], product operation [1,25], and performance [1,24,26]. These supplier-level studies primarily show the advantages of professional suppliers in resources and capabilities. Furthermore, some researchers also from the platform perspective investigate the relative impacts of such two types of suppliers on platform quality [45,49] and performance [2,16,23,50,51]. These platform-level studies suggest that professional suppliers (vs. nonprofessional suppliers) play a more vital role in platform performance and quality.

This study differs from prior studies in two respects. First, while extant literature reveals the impacts of different suppliers on platform quality and performance, little is known about their impacts on platform growth. Given the results of heterogeneous CNEs driven by different suppliers, our study contributes by providing insights into their relative roles in platform growth. Second, although most studies demonstrate the superiority of professional suppliers, they fail to account for the presence of nonprofessional suppliers. This study contributes to the literature by revealing the superiority of nonprofessional suppliers, which extends the theoretical understanding of the two types of suppliers.

As illustrated in Fig. 1, we integrate the two streams of literature directly related to this study: the literature on heterogeneous CNEs and the literature on two types of suppliers. This synthesis reveals a primary research gap in the overlapping area, namely, the limited understanding of the relative impacts of nonprofessional and professional suppliers on CNEs. Therefore, to the best of our knowledge, this study is the first to empirically examine the relative roles of nonprofessional and professional suppliers in stimulating CNEs, which contributes primarily by enhancing the theoretical understanding of both CNEs and the dynamics of the two types of suppliers.

3. Theoretical development and hypotheses

3.1. Differences between nonprofessional and professional suppliers

Extant literature provides a well-established understanding of nonprofessional and professional suppliers. Specifically, as shown in Table 1, we summarize their differences into three attributes: identity formality, resource richness, and entry motivation.

Identity formality refers to a supplier's degree of formality in his/her job and industry. Nonprofessional suppliers are generally small teams or amateur individuals operating on a part-time and flexible basis [46]. By contrast, professional suppliers are specialized firms or full-time workers dedicated to their jobs [47,2,25]. For example, most large sellers (i.e., professional suppliers) on eBay operate dedicated firms, while small sellers (i.e., nonprofessional suppliers) are typically individuals or small teams who relatively occasionally do business on the platform [45]. Therefore, we consider that professional suppliers are likely to have a formal identity, whereas nonprofessional suppliers tend to have an informal identity.

Resource richness refers to the richness of tangible resources (e.g., products, money) and intangible resources (e.g., time, particular skills) that a supplier uses for business. Existing literature indicates that professional suppliers generally possess richer resources than nonprofessional suppliers [1,25,26]. For example, professional hosts on Airbnb are

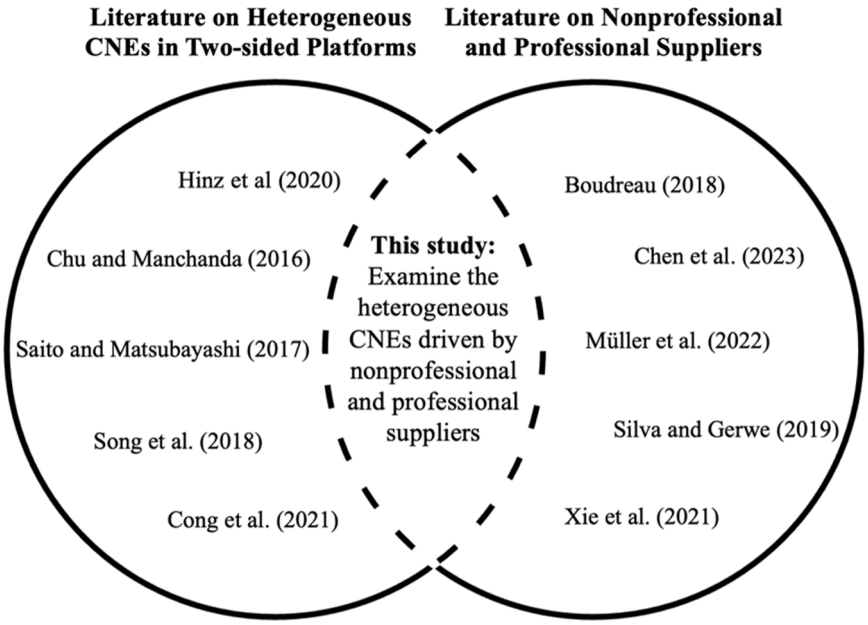


Fig. 1. Research gap in platform literature.

Table 1
Differences of nonprofessional and professional suppliers on P2P platforms.

Attribute	Nonprofessional Supplier	Professional Supplier	Related Studies
Identity formality	No formal identity, e.g., a flexibly operated small team or a part-time worker	Formal identity, e.g., a specialized firm or a full-time worker	Boto-García et al. [47]; Einav et al. [2]; Müller et al. [25]
Resource richness	Limited resources like time, specialist skills, etc.	Rich resources like specialist skills, time, market information, etc.	Casamatta et al. [48]; Chen et al. [1]; Xie et al. [26]; Müller et al. [25]
Entry motivation	Motivated by the minimum supply costs	Motivated by the maximum profits	Boudreau [45]; Li et al. [23]; Silva and Gerwe [46]

likely to operate much more properties and perform better than nonprofessional hosts [47,48,1]. Similarly, large or professional sellers on eBay have more market information and richer marketing experiences than small sellers do.

Entry motivation refers to the primary factor influencing the entry behavior of suppliers. Professional suppliers are mainly motivated by the maximum profits, while nonprofessional suppliers are mainly motivated by the minimum costs [45,2,46]. For example, nonprofessional hosts on Airbnb are primarily motivated by the low entry barriers and the opportunity to earn extra income. By contrast, professional hosts are motivated by the pursuit of profit maximization [1,23]. Similarly, some nonprofessional sellers on eBay are motivated to earn extra income by selling second-hand or handmade products.

Based on these three attributes of the two types of suppliers, we define *nonprofessional suppliers* as those who have an informal identity, limited resources, and are motivated by the minimum supply costs. *Professional suppliers* are defined as profit-seekers with a formal identity and rich resources (e.g., time, money, particular skills). *Identity formality* serves as a fundamental qualitative attribute that inherently distinguishes behavioral patterns between the two types of suppliers. *Resource richness* and *entry motivation* represent quantitative yet latent attributes that further delineate the extent of behavioral divergence between these supplier types. Importantly, we consider that these three attributes operate in a complementary manner and collectively shape the behavioral dynamics between nonprofessional and professional suppliers, which helps explore the heterogeneous CNEs between different suppliers and consumers.

3.2. The value of CNEs: quantity and quality effects

As defined earlier, the strength of CNEs depends on how one side's characteristics impact the other side's benefits [11]. Quantity is a well-established characteristic on both the supply and demand sides, driving CNEs by influencing the availability and variety of mutual interactions [36,10,19]. In addition to quantity, an increasing number of studies suggest that quality is another critical but understudied characteristic on the supply side, which would influence the consumption benefits of consumers, thereby influencing the strength of CNEs [5,50,52,53]. Therefore, we consider that utilizing both quantity and quality on the supply side could help better capture CNEs.

Following prior literature, *supply quantity* refers to the number of suppliers or products on the supply side, which relates to the set of opportunities available to consumers for transacting and, thus, the likelihood of meeting the different needs and preferences of consumers [5,6]. *Supply quality* refers to the quality of products and services on the supply side, which relates to the benefits that consumers derive from interacting and transacting with suppliers [54,15]. Building on the two characteristics, we refer to *quantity effects of CNEs* as the mutual influences between supply quantity and consumer size, and refer to *quality effects of CNEs* as the mutual influences between supply quality and consumer size.

3.3. The quantity effects of CNEs between suppliers and consumers

An increase in the quantity of nonprofessional and professional suppliers typically results in varying levels of supply diversity on platforms, which in turn leads to differentiated impacts on consumer acquisition and growth. The product scope of nonprofessional suppliers tends to be narrow and specialized due to resource limitations [55,45]. For example, small sellers on eBay often provide niche products and unique services [2]. Similarly, nonprofessional hosts in our data generally operate properties scattered across various districts. Therefore, adding a number of nonprofessional suppliers is considered to increase the supply diversity on the platform, helping attract consumers by meeting heterogeneous needs and tastes.

By contrast, the product scope of professional suppliers tends to be wide and general, as such an operation could help them achieve economies of scope and maximize profits [56,44]. Though some of them might contemplate niche markets, they may soon give up because these markets can't sustain their growth needs, making economies of scale hard to achieve [46]. For example, either on Airbnb or in our empirical data, the properties owned by professional hosts are often clustered in economically successful districts near the city center, subway stations, or universities [1,57]. Adding a number of professional suppliers thus could help platforms attract consumers by meeting general needs and tastes. Therefore, we consider that adding different suppliers would result in different levels of supply diversity and different attractiveness to consumers.

Given the importance of supply diversity in driving consumer growth, which has been persuasively argued in numerous platform contexts [36,2,58], we consider that adding a number of nonprofessional suppliers (vs. professional suppliers) would better help platforms drive consumer growth by expanding the diversity of products available on platforms and meeting more heterogeneous needs. This is aligned with some practical observations. For example, nonprofessional hosts are considered to play a major role in the growth of Airbnb's accommodation business. Similarly, aiming to "make it easy to do business anywhere," Taobao has rapidly grown by allowing individual sellers to freely open online stores [20].

Furthermore, adding a number of nonprofessional suppliers (vs. professional suppliers) would decrease the average product price on the platforms, thereby creating more consumer surplus and attracting more consumers [2,6]. Setting competitive prices needs rich market information and experience, which is relatively cost-effective for professional suppliers but costly for nonprofessional suppliers [47,44]. For example, the properties owned by professional hosts on Airbnb are found to be priced higher than those of nonprofessional hosts [1]. This is plausible because professional suppliers have richer resources and skills than nonprofessional suppliers. Thus, given the above potential impacts of adding different suppliers, we consider that adding a number of nonprofessional suppliers (vs. professional suppliers) could generally help platforms attract more consumers (i.e., stimulate stronger CNEs). Here, we propose hypothesis 1a (H1a):

H1a. *In the quantity effects of CNEs, nonprofessional-to-consumer CNEs are stronger than professional-to-consumer CNEs.*

In turn, regarding consumer-to-supplier CNEs, adding a number of consumers would drive the growth of nonprofessional and professional suppliers differently by creating different values. When consumer size increases, the heterogeneity of consumers' needs on platforms increases on average [2,58], thereby providing suppliers with more diverse opportunities. In this way, nonprofessional suppliers characterized by specialized product scopes, would be attracted to meet the heterogeneous needs. For example, the increasing size of buyers on eBay attracts a number of amateur sellers and results in various interesting and niche products (e.g., vintage treasures, obscure collections). By contrast, while adding consumers also creates more potential value for professional suppliers, it's costly for them to flexibly allocate resources in response to the increased heterogeneous needs [46]. Thus, adding consumers seems to have a larger attractiveness to nonprofessional suppliers (vs. professional suppliers) by providing larger value.

Furthermore, according to the 20/80 principle, also known as the Pareto principle, 20 % of consumers generally account for 80 % of sales volume on the market [59,60]. In this way, professional suppliers are likely to focus on the most profitable consumers (i.e., 20 % of consumers to maximize profits), which is also aligned with their wide and general product scopes. By contrast, in alignment with narrow and specialized product scopes, nonprofessional suppliers are likely to focus on the remaining 80 % of consumers to capture potential heterogeneous value. In sum, adding a number of consumers is likely to help platforms attract more nonprofessional suppliers than professional suppliers. That is, in consumer-to-supplier CNEs, the consumer size has stronger CNEs on the quantity of nonprofessional suppliers (vs. professional suppliers). Here, we propose hypothesis 1b (H1b):

H1b. *In the quantity effects of CNEs, consumer-to-nonprofessional CNEs are stronger than consumer-to-professional CNEs.*

3.4. The quality effects of CNEs between suppliers and consumers

The quality enhancement of nonprofessional and professional suppliers would create different benefits for consumers, thereby resulting in different impacts on consumer acquisition and growth. Generally, the basic quality of suppliers depends on the extent to which suppliers satisfy consumers [61,62]. In other words, quality enhancement could help suppliers improve their ability to satisfy consumers. Thus, for nonprofessional suppliers, enhancing quality could help platforms improve the uniqueness and attractiveness of products, thereby better meeting the needs of niche markets. For example, individual sellers on eBay satisfy buyers by offering unique products and specialized services [2]. By contrast, for professional suppliers, enhancing quality could help platforms better meet the general market demand. For example, professional hosts on Airbnb would provide hotel-like properties and standardized accommodation experiences to attract and satisfy renters [1]. Therefore, given the characterization of heterogeneous demands on platforms (as argued in H1), we consider that the quality enhancement of nonprofessional suppliers (vs. professional suppliers) could better help platforms drive consumer growth by more effectively satisfying heterogeneous needs.

The quality enhancement of different suppliers would also create different marginal benefits for consumers, leading to varying impacts on consumer growth. It's acknowledged that the quality of professional suppliers is on average higher than that of nonprofessional suppliers, as professional suppliers have richer experience and skills [45,48]. In other words, consumers are likely to have higher expectations of the quality of professional suppliers (vs. nonprofessional suppliers). The higher the consumer expectation, the harder it is for suppliers to satisfy consumers by achieving or even exceeding expectations [63]. Therefore, the marginal benefits of quality enhancement of professional suppliers seem to be less than those of nonprofessional suppliers because it's harder for them to achieve or exceed consumer expectations. In sum, we consider that the quality enhancement of nonprofessional suppliers (vs. professional suppliers) could better drive consumer growth by increasing consumer benefits on the platforms. Here, we propose the following:

H2a. *In the quality effects of CNEs, nonprofessional-to-consumer CNEs are stronger than professional-to-consumer CNEs.*

In turn, regarding consumer-to-supplier CNEs, adding consumers could help facilitate the quality enhancement of suppliers, as suppliers are motivated to improve quality to acquire and satisfy consumers [64]. These impacts may vary for professional and nonprofessional suppliers due to their differences in resource richness. Theoretically and practically, quality enhancement is a resource commitment process [65,19]. Professional suppliers who have richer resources than nonprofessional suppliers could better invest time, money, and effort in enhancing quality to satisfy consumers [47,1]. By contrast, though some nonprofessional suppliers may also be motivated to improve quality, it's relatively costly for them to invest lots of resources due to resource limitations [45]. This pattern could be evident in several practical examples. For example, professional hosts on Airbnb generally perform better in satisfying renters than nonprofessional hosts do [1]. Some individual or flexible sellers on eBay are reluctant to invest abundant time and effort in interacting with buyers. Thus, we consider that with an increase in consumer size, the quality enhancement of professional suppliers could increase more than that of nonprofessional suppliers. Here, we propose the following:

H2b. *In the quality effects of CNEs, consumer-to-professional CNEs are stronger than consumer-to-nonprofessional CNEs.*

4. Methodology

4.1. Research context

We collaborated with a P2P platform of real estate rental properties in China. Due to the confidentiality agreement with the platform, we do not disclose the name of the platform. The platform was established in 2015 and has become one of the major rental platforms in China. It builds an app and a website for both renters and hosts to communicate and transact with each other. Its business model is similar to that of Airbnb. It provides a marketplace for hosts to list their rental properties with some attributes (e.g., pictures, location, size, amenities, rent), and for prospective renters to discover and rent properties. Different from Airbnb and some other platforms that mainly facilitate short-term rental transactions, the focal platform focuses on relatively long-term rental periods that usually span at least one month.

4.2. Empirical setting: identification of different suppliers

In our empirical context, the platform has distinct labels on the supply side (i.e., host side), which provides us with a unique opportunity to investigate the implications and impacts of different suppliers (i.e., hosts in this study). Specifically, the platform distinguishes hosts with three labels according to their registered identity information: nonprofessional host, housing rental agent, and real estate agent. Hosts labeled as "nonprofessional host" are homeowners who are allowed to list three properties every month for free on the platform by uploading fundamental information (e.g., personal and housing information). Hosts labeled as of "housing rental agent" are experts hired by homeowners to assist in renting out properties by marketing, searching for renters, and interacting with renters. Once the rental contract is signed, their work is finished. Hosts labeled as of "real estate agent" are employees of real estate agencies who rent properties from homeowners and sublet them to renters after making some standardized renovations to the properties [66]. They handle all aspects of property management, such as searching for renters, setting rental prices, and collecting rent. Different from

nonprofessional hosts, the latter two kinds of hosts need to register company information, and they need to pay entry fees to list properties without quantity limits. Notably, the three types of hosts are individual service providers with different identities on the platform.

Uniquely, based on the given labels on the supply side, we could define *nonprofessional hosts* as those individual amateurs with the label of "nonprofessional host" on the platform, and define *professional hosts* as those experienced intermediate hosts with the label of "housing rental agent" or "real estate agent" because they both have expertise in operating properties. Indeed, we consider that such definitions and classifications are consistent with the theoretical understanding of the two types of suppliers mentioned before. Specifically, most professional hosts are full-time workers and profit-seekers coming from firms like housing rental agencies or real estate agencies, thus, each of them has a formal identity and is motivated by maximum profits. Also, they tend to have rich resources like time, skills, and market information. By contrast, most nonprofessional hosts operate on a part-time basis, characterized by limited resources and informal operational identities, with their participation primarily motivated by monetizing the underutilized properties.

4.3. Data and measurement

We collect data from January 1, 2016, to December 26, 2019, in five major Chinese cities. Indeed, the observation window is set based on two considerations. First, the platform was established in late 2015 and only entered one city in 2015, resulting in sparse data from the five cities. As data from the five cities has been rich since 2016, we consider it proper to collect data from 2016. Moreover, the entry and usage behaviors of prospective users (i.e., hosts and renters) may be influenced by the COVID-19 pandemic that occurred at the beginning of 2020. We acknowledge that such impacts on the focal platform are exogenous and can't be ignored. Thus, we collected data up to 2019 to exclude the exogenous shock of the COVID-19 pandemic. Second, we consider the three-year observation window to be sufficient to capture the mutual influences among focal variables. Indeed, our empirical results suggest that the mutual CNEs between suppliers and consumers not only emerge within 20 weeks but also tend to stabilize by the 20th week.

During our observation period in five major cities, there are 1370,007 registered renters, 1470,184 hosts listing properties (38 % are professional hosts and 62 % are nonprofessional hosts), and more than two million properties. Specifically, the data set includes property profile information (e.g., host information, location, rent, number of rooms, rental status), temporal interaction logs (e.g., call duration timestamps, contact frequency, conversation content), and renter complaint records (e.g., complaint reasons and times). We test our hypotheses using a final panel district-week-level dataset of 894 districts over 208 weeks. The data aggregated at the district level allows us to control for heterogeneity across different districts and leverage the growth fluctuations within these districts to explore CNEs between hosts and renters. In addition, the data is aggregated at the weekly level mainly because daily data on CNEs does not entail sufficient variations, and monthly data does not provide the granularity to reveal ongoing patterns of CNEs. Table 2 summarizes all measures and their descriptive statistics.

4.3.1. Supply quantity

Following extant studies, we use the number of active suppliers (i.e., hosts) to measure the supply quantity and reflect the CNEs on the supply side [20,10,31,12]. We consider active hosts as those who have listed at least one property in district *i*. If a host has listed properties in more than one district, he/she will be treated as an active host in each of these districts. In this way, the measures of the supply quantity of two different hosts, $NonproQuantity_{it}$ and $ProQuantity_{it}$, are calculated as the total number of active nonprofessional hosts and professional hosts in district i and week t, respectively. We use the natural logarithm of $NonproQuantity_{it}$ and $ProQuantity_{it}$ values to address the skew in distributions.

Table 2
Variables, measurement, and descriptive statistics.

Variables	Measurement	Mean	SD	Min	Max
Focal Variables					
NonproQuantity	Total quantity of nonprofessional hosts in district *i* in week *t* (log)	5.358	1.937	0.000	9.214
ProQuantity	Total quantity of professional hosts in district *i* in week *t* (log)	4.602	2.213	0.000	8.828
NonproQuality	1 - ratio of the number of complained nonprofessional hosts' properties to the total number of nonprofessional hosts' properties in district *i* in week *t*	1.000	0.009	0.000	1.000
ProQuality	1 - ratio of the number of complained professional hosts' properties to the total number of professional hosts' properties in district *i* in week *t*	0.999	0.013	0.000	1.000
RenterSize	Total number of renters in district *i* in week *t* (log)	5.116	1.980	0.000	9.188
Control Variables					
MatchRate	Ratio of rented-out properties to the total number of properties in district *i* in week *t*	0.021	0.060	0.000	1.000
MatchDay	(Average days required to rent out a property in district *i* in week *t*)/100	0.466	0.647	0.000	12.340
HostCoverage	Average number of districts covered by hosts in district *i* in week *t*	2.475	4.905	1.000	245.000
RenterCoverage	Average number of districts covered by renters in district *i* in week *t*	4.092	2.115	1.000	46.000
AvgRent	(Average rent per square meter of properties in district *i* in week *t*)/100	0.814	0.377	0.069	6.656
Holiday	Dummy (1: week *t* includes legal holiday, 0: otherwise)	0.172	0.377	0	1

Note: Number of observations: 164,979.

4.3.2. Renter size

Following extant empirical studies (e.g., [19]), we use the number of active renters to measure the renter size. As calling hosts is a very common and necessary interaction for prospective renters and hosts [67], it's reasonable to identify active renters based on the calling behaviors. Therefore, we define active renters in a district as those who have made at least one phone call to hosts in the district. If a renter has called hosts in more than one district, he/she will be considered as an active renter in each of these districts. Then the measure of renter size, $RenterSize_{it}$, is calculated as the total number of active renters in district i in week t. We use the natural logarithm of $RenterSize_{it}$ values to address the skew in distributions.

4.3.3. Supply quality

The measurement of supply quality is challenging as quality is a complex and multidimensional construct [54]. We use complaint data to indirectly capture supply quality for the following reasons. First, theoretically, consumer complaints are a kind of consumer feedback that

could be used to capture the quality of suppliers, in addition to online reviews and ratings [68-71]. On the one hand, consumer complaints are closely related to the product and service quality of suppliers, especially in the service industry, as consumers mainly perceive service quality when they experience problems [65,72-74]. High-quality suppliers (vs. low-quality suppliers) usually provide more satisfying products and result in a lower likelihood of consumer complaints [75,74]. Similarly, in our empirical context, high-quality hosts (vs. low-quality hosts) are less likely to be complained about by renters because they are likely to offer satisfying properties and services. On the other hand, consumer complaints contain valuable information like the causes of service failure, dissatisfaction, or other opinions [64,76,77]. Both platforms and suppliers could utilize consumer complaint data to obtain useful feedback for quality improvements to increase consumer satisfaction [65, 78-80].

Second, the focal platform primarily monitors host quality based on renters' complaint behaviors rather than reviews or ratings. This indicates the important role of consumer complaints in capturing suppliers' quality in practice. Specifically, the focal platform has built a complaint feedback channel, allowing renters to lodge official complaints about hosts. On the page of each listed property, there is a link for renters to submit a complaint about their dissatisfaction with the property or the host's service (see Appendix A). Once a complaint is submitted, platform managers will contact the corresponding host to verify whether the complaint is valid. Usually, renters will receive a response about the complaint from the platform within 24 h. If the complaint is valid, the platform will remove the listing of the involved property from the website for a month and record the verified complaint rate. If a host has accumulated three verified complaints, the platform will suspend the host's account for three months. If the complaint is invalidated, the renter will be recorded by the platform. If a renter has accumulated more than three invalid complaints in a month, the renter will be blocked from contacting and renting properties on the platform in the future.

Therefore, given these theoretical and practical roles of consumer complaints in capturing suppliers' quality, we employ complaint data to derive measures of supply quality as follows. First, we calculate N_{it}^{nonpro} that measures the proportion of the number of complained nonprofessional hosts' properties in the total number of nonprofessional hosts' properties in district i and week t. Notably, if a property has been complained about more than once in the same week, we only count it once. Similarly, we calculate N_{it}^{pro} that measures the proportion of the number of complained professional hosts' properties in the total number of professional hosts' properties in district i in week t. A lower level of N_{it}^{nonpro} or N_{it}^{pro} suggests a higher degree of host quality. That is, we obtain two negative indicators. Second, to obtain positive indicators and make the interpretation more intuitive, we create two variables—$NonproQuality_{it}$ and $ProQuality_{it}$. We let $NonproQualityit$ equal 1-N_{it}^{nonpro} and $ProQuality_{it}$ equal 1-N_{it}^{pro}, where 1 is the maximum value *of* N_{it}^{nonpro}(N_{it}^{pro}). In this way, a higher value of $NonproQuality_{it}$ ($ProQuality_{it}$) means a higher level of quality of nonprofessional hosts (professional hosts) in district i in week t.

4.3.4. Control variables

We also control for several district-level and time-variant characteristics that may affect the growth of the two sides. First, we construct $MatchRate_{it}$ measured by the ratio of the rented-out properties to the total number of properties in district i in week t, which could indirectly affect the mutual influences between hosts and renters. For example, the districts with higher matching rates are more likely to be attractive and popular to both hosts and renters, even though the host size and renter size are small. Second, we construct $MatchDay_{it}$ measured by the average days required to rent out a property in district i in week t. The average time spent to make a deal is relevant to the transaction cost of both hosts and renters. The districts with lower $MatchDay_{it}$ are likely to attract both hosts and renters to enter the markets. Third, as some renters and hosts may be active in more than one district, the same user (either renter or host) may be counted repeatedly at the platform level and resulting in the inflation of user size. To control such inflation effects, we construct two variables, $HostCoverage_{it}$ and $RenterCoverage_{it}$, measured by the average number of districts covered by hosts and renters in district i in week t, respectively. Fourth, it's wise to control the impacts of the average rent in a district, because both renters and hosts are generally sensitive to rent. Therefore, we construct $AvgRent_{it}$ measured by the average rent per square meter in district i in week t. Finally, we construct $Holiday_t$, a dummy variable that measures whether week t includes any legal holiday. The week with or without any legal holiday may influence the growth of renters. We summarize all variables and their descriptive statistics in Table 2.

4.4. Empirical approach and model specification

The identification and measurement of CNEs has been regarded as the most challenging, as the mutual influences between the two sides are dynamic and complicated [53]. Following extant literature, we identify CNEs by examining how much one side's attribute changes in response to a change in the other side's attribute [18,19]. Specifically, host-to-renter CNEs are identified by examining how the change in attribute (either quantity or quality) on the supply side results in a change in renter size. Renter-to-host CNEs are identified by examining how the change in renter size results in a change in attribute (either quantity or quality) on the supply side.

To capture these effects, we employ a PVAR analysis with generalized method of moments (GMM) estimation, which has the following advantages in our empirical context. First, this approach doesn't require prior information and assumes all variables are endogenous, which can not only address the endogeneity issue but also simultaneously capture the dynamic and intricate bidirectional relationships among different variables [81,82]. It therefore allows us to characterize host-to-renter CNEs and renter-to-host CNEs based on host type and host-side attributes (i.e., quantity and quality). Second, the PVAR model can simulate how the change in one variable (e.g., an increase of host quantity) has a dynamic impact on another variable (e.g., renter size) through Impulse Response Functions (IRFs). This allows us to identify the magnitude of CNEs and helps better illustrate CNEs [83-85]. Third, the PVAR model can handle unobserved individual heterogeneity and account for cross-sectional dynamic heterogeneities [86,87]. It therefore helps control for unobserved district heterogeneity and obtain consistent results.

Following classic PVAR methodological literature (e.g., [86,87,81]) and recent empirical studies published in top journals (e.g., [83,88,82]), we employ the PVAR model by the following four steps. First, analyze the endogeneity of focal variables based on Granger causality tests. Second, evaluate the stationarity of focal variables based on unit-root tests. Third, select the optimal lag terms through information criteria to construct the PVAR model. Fourth, test hypotheses using cumulative orthogonalized IRFs.

4.4.1. Endogeneity test

To employ PVAR to capture the influences among the endogenous variables, we need to first examine whether the focal variables are endogenous. Empirically, the Granger causality test is widely used to establish causality in time series data [89]. Consistent with prior studies [84,19], we conduct a series of Granger causality tests, starting with 1 lag and stopping at 20 lags, to explore whether explanatory variables explain the variation of dependent variables, in addition to the lagged values of dependent variables. As summarized in Table 3, Granger causality is established for all pairs.

4.4.2. Stationarity test

GMM estimators suffer from the problem of the weak instrument if endogenous variables contain unit roots [86]. Thus, we need to examine

Table 3
Results of Granger causality tests.

Independent Variable	Dependent Variable				
	NonproQuantity	*ProQuantity*	*NonproQuality*	*ProQuality*	*RenterSize*
NonproQuantity	—	0.002**	0.001**	0.000***	0.000***
ProQuantity	0.000***	—	0.002**	0.050*	0.000***
NonproQuality	0.000***	0.000***	—	0.000**	0.000***
ProQuality	0.002***	0.000***	0.000***	—	0.000***
RenterSize	0.000***	0.000***	0.000***	0.007**	—

Notes: Tests were done with up to 20 lags, and we report the smallest p-values of the chi-square tests across the models with 20 different lags. Significance levels: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

the stationarity assumption of the five endogenous variables using the unit-root tests. Because our panel data is unbalanced, certain popular panel unit-root tests, such as the Levin–Lin–Chu test and ADF test, are inappropriate, as they require strongly balanced panel data. We thus resort to the Fisher-type approach for testing unit roots in panel data [84]. To remove panel-specific fixed effects, we use first-difference values in the tests. As shown in Table 4, the null hypothesis of having a unit root is rejected for all five variables, suggesting that all variables are stationary.

4.4.3. Model specification with optimal lag

According to extant literature [86,90], the presence of endogeneity and stationarity in all the variables suggests that applying the panel VAR model rather than the panel VECM model is more appropriate. Thus, our PVAR model is as follows:

$$\begin{bmatrix} NonproQuantity_{i,t} \\ ProQuantity_{i,t} \\ NonproQuality_{i,t} \\ ProQuality_{i,t} \\ RenterSize_{i,t} \end{bmatrix} = \sum_{p=1}^{p} \begin{bmatrix} \beta_{11}^{p} & \beta_{12}^{p} & \beta_{13}^{p} & \beta_{14}^{p} & \beta_{15}^{p} \\ \beta_{21}^{p} & \beta_{22}^{p} & \beta_{23}^{p} & \beta_{24}^{p} & \beta_{25}^{p} \\ \beta_{31}^{p} & \beta_{32}^{p} & \beta_{33}^{p} & \beta_{35}^{p} & \beta_{35}^{p} \\ \beta_{41}^{p} & \beta_{42}^{p} & \beta_{43}^{p} & \beta_{44}^{p} & \beta_{45}^{p} \\ \beta_{51}^{p} & \beta_{52}^{p} & \beta_{53}^{p} & \beta_{54}^{p} & \beta_{55}^{p} \end{bmatrix} \begin{bmatrix} NonproQuantity_{i,t-p} \\ ProQuantity_{i,t-p} \\ NonproQuality_{i,t-p} \\ ProQuality_{i,t-p} \\ RenterSize_{i,t-p} \end{bmatrix} + \sum_{p=1}^{p} \begin{bmatrix} \varphi_{11}^{p} & \cdots & \varphi_{16}^{p} \\ \varphi_{21}^{p} & & \varphi_{26}^{p} \\ \varphi_{31}^{p} & \ddots & \varphi_{36}^{p} \\ \varphi_{41}^{p} & & \varphi_{46}^{p} \\ \varphi_{51}^{p} & \cdots & \varphi_{56}^{p} \end{bmatrix} \begin{bmatrix} MatchRate_{i,t-p} \\ MatchDay_{i,t-p} \\ AvgRent_{i,t-p} \\ Holiday_{i,t-p} \\ HostCoverage_{i,t-p} \\ RenterCoverage_{i,t-p} \end{bmatrix} + \mu_i + \varepsilon_{it} \quad (1)$$

In this model, the coefficient matrix of β denotes the parameters to be estimated that capture the effects of the lagged endogenous variables; p indicates the number of lagged periods; the coefficient matrix of φ measures the effects of all exogenous and control variables; μ_i captures the panel fixed effects, and ε_{it} is a vector of the idiosyncratic error terms that are assumed to be zero-centered.

To complete the model, we need to determine the appropriate number of lags, i.e., the parameter p in the model. We thus use Andrews and Lu's [91] moment and model selection criteria to select a proper one, which include various information criteria, such as Akaike information criterion (AIC), Bayesian information criterion (BIC), and Hannan-Quinn information criterion (HQIC). Ideally, the optimal lag is the one that minimizes AIC, BIC, and HQIC simultaneously. In our model, both the AIC and HQIC suggest that the lag of two periods is the optimal choice (AIC = −122.32, HQIC = −481.35). Thus, our PVAR model includes lags of up to two weeks.

Table 4
Results of panel unit root tests.

Variable	Inverse Chi-Squared	*P*-Value	Conclusion
NonproQuantity	59,300	0.000	Stationary
ProQuantity	57,100	0.000	Stationary
NonproQuality	63,000	0.000	Stationary
ProQuality	62,100	0.000	Stationary
RenterSize	57,800	0.000	Stationary

Notes: Number of panels = 894; H_0: All panels contain unit roots; H_a: At least one panel is stationary.

4.4.4. Impulse response function

After completing the model specification with an optimal lag of two, we derive the coefficient estimations of the main variables from PVAR model. Then we use the estimations to construct the impulsive response functions, i.e., IRFs, to test our hypotheses. We choose IRFs for two reasons. First, according to extant literature [83,84], the estimated PVAR coefficients are not sufficiently informative to establish causal relationships among variables due to the complicated dynamics in inherent PVAR systems. Second, IRFs can illustrate how the shock of one standard deviation on one variable affects another variable over time, with other variables in the system remaining unchanged [90,82,85]. That is, it's appropriate to use IRFs to portray the causal effects of bidirectional relationships. Hence, following recent empirical studies (e.g., [83,84,82]), we rely on the post-estimation orthogonalized IRFs to test our hypotheses (in Section 5.2).

5. Results

5.1. Model-free evidence: differences between nonprofessional and professional hosts

To illustrate the potential differences between nonprofessional and professional hosts in terms of identity formality, resource richness, and entry motivation, we compare the property and service characteristics of different hosts using data aggregated at the host level. As summarized in Table 5, the descriptive statistic results are consistent with our theoretical understanding of the two types of suppliers in Section 3.1. First, as clarified in Section 4.2, hosts differ clearly in terms of identity formality, as the platform explicitly classifies them as either nonprofessional or professional based on assigned identity labels.

Second, both the property and service characteristics in Table 5 demonstrate hosts' differences in resource richness, indicating that professional hosts have richer resources than nonprofessional hosts. Specifically, professional hosts (vs. nonprofessional hosts) have more properties (Difference = 0.779, $p < 0.001$) with more bedrooms (Difference = 0.286, $p < 0.001$) and larger sizes (Difference = 42.590, $p < 0.001$). The ratio of entire properties of professional hosts is also higher (Difference = 0.514, $p < 0.001$). Furthermore, professional hosts (vs. nonprofessional hosts) on average commit more time and effort in

Table 5
Comparisons of nonprofessional and professional hosts.

Attribute	Measurement	Nonprofessional Hosts		Professional Hosts		Difference
		Mean	SD	Mean	SD	
Property characteristics						
#Property	Number of properties	1.291	1.441	2.071	13.962	0.779***
#Bedrooms	Average number of bedrooms per property	2.224	0.921	2.510	1.070	0.286***
Size	Average size per property	35.591	30.287	78.181	44.011	42.590***
%Entire	Ratio of entire properties	0.382	0.482	0.896	0.298	0.514***
Rent	Average rent per property	2360.294	1804.430	6743.304	5814.393	4383.010***
Distance	Average kilometers from the located district to the city center per property	12.919	8.377	9.827	7.145	−3.092***
Service and marketing characteristics						
Response rate %	The number of connected calls divided by the total number of incoming calls	0.701	0.391	0.745	0.346	0.044***
Call duration	Total minutes for connected calls	4.214	5.635	11.020	30.152	6.806***
Product scope	Number of unique property categories	1.092	0.364	1.274	0.775	0.182***
Quality	1-ratio of complained properties	0.989	0.102	0.996	0.055	0.007***
Number of observations		906,726		563,458		

Notes: *$p < 0.01$; **$p < 0.05$; ***$p < 0.001$.

operation, i.e., higher response rate (Difference = 0.044, $p < 0.001$) and longer call duration (Difference = 6.806, $p < 0.001$). These suggestive results are also consistent with our arguments in H2b.

Third, the results in Table 5 also indicate hosts' differences in entry motivation. Specifically, the average rent per property of professional hosts is higher than that of nonprofessional hosts (Difference = 4383.010, $p < 0.001$), which implies that professional hosts are profit-seekers motivated by profits. In addition, the properties of professional hosts (vs. nonprofessional hosts) are on average closer to the city center (Difference = −3.092, $p < 0.001$), which indicates professional hosts would like to target popular markets to maximize profits. This also aligns with our arguments in H1a and H2a. In sum, these descriptive statistical results exhibit a consistent pattern with our theoretical understanding of professional and nonprofessional suppliers.

Furthermore, we also create two attributes: *Product scope* and *Quality,* as shown in Table 5, to demonstrate the behavioral differences argued in our hypotheses. First, we identify five key features of properties and classify each feature into two or three categories: the number of bedrooms (high/medium/low), size (large/medium/small), rent (high/medium/low), distance to the city center (far/moderate/near), and rental type (entire/shared). Among these features, the rental type is divided into two categories based on a dummy variable in host listing data, while each of the other features is divided into three categories based on the quantiles of related continuous variables. By combining the categories of five features, the properties on the platform are classified into 162 unique categories (i.e., 3 × 3 × 3 × 3 × 2). Then, *Product scope* is operationalized as the number of unique property categories owned by a host. Consistent with our arguments in H1 and H2, the product scope of professional hosts is on average wider than that of nonprofessional hosts (Difference = 0.182, $p < 0.001$). Second, we let *Quality* = 1-ratio of complained properties. Thus, a higher value of *Quality* means a higher quality of the host. Table 5 shows that the quality of professional hosts is on average higher than nonprofessional hosts (Difference = 0.007, $p < 0.001$), which is aligned with our arguments in H2b.

5.2. Results of hypothesis testing

Applying the PVAR model, we obtain the estimations of PVAR coefficients, and we use the estimates to derive IRFs to test our hypotheses. Specifically, using the orthogonalized IRFs derived from the estimation of the PVAR model, we simulate the fitted PVAR system using a Monte Carlo method with 1000 runs. Then, we calculate the cumulative orthogonalized IRFs as shown in Table 6 and use a time window of 20 weeks to illustrate the IRFs in Fig. 2. Finally, as we mainly focus on the differential effects, we calculate the differences among estimated IRFs and test their significance to accomplish hypothesis testing (see Table 6).

Table 6
Cumulative IRFs of CNEs between hosts and renters.

Panel A: Host-to-Renter CNEs			
	Host Type		Difference
Paths	Nonprofessional	Professional	
Supply quantity → Renter size (H1a)	0.0264**	0.0035*	0.0230***
	(0.002)	(0.001)	(0.003)
Supply quality → Renter size (H2a)	0.0183***	−0.0123***	0.0306***
	(0.004)	(0.001)	(0.005)
Panel B: Renter-to-Host CNEs			
	Host Type		Difference
Paths	Nonprofessional	Professional	
Renter size → Supply quantity (H1b)	0.0326***	0.0160***	0.0166***
	(0.001)	(0.002)	(0.002)
Renter size → Supply quality (H2b)	−0.0001**	0.0003*	−0.0004***
	(0.000)	(0.000)	(0.000)

Notes: The table presents cumulative orthogonalized IRFs at week 20. Robust standard errors are presented in parentheses. Significance levels: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

5.2.1. Heterogeneous CNEs between supply quantity and renter size

Table 6 reports the hypothesis test results of the heterogeneous CNEs between two types of hosts and renters. Besides, Fig. 2 graphically shows the dynamic relationships between the host side and the renter side, in which the solid lines with triangles represent nonprofessional hosts and the dotted lines with circles represent professional hosts. Regarding host-to-renter CNEs, results in Table 6 show that the effects of *supply quantity* on *renter size* significantly differ depending on the host type (either professional hosts or nonprofessional hosts). Specifically, the *supply quantity* of nonprofessional hosts has a larger effect on *renter size* than the *supply quantity* of professional hosts does (Difference = 0.0230, $p < 0.001$). Fig. 2(a) demonstrates that the response of renter size to an increase in supply quantity of nonprofessional suppliers (vs. professional suppliers) is larger. These results suggest that adding a number of nonprofessional hosts (vs. professional hosts) could better attract renters. That is, in the quantity effects of CNEs, the nonprofessional-to-renter CNEs are stronger than the professional-to-renter CNEs. Hence, H1a is supported.

In turn, in the renter-to-host CNEs, Table 6 shows that the effects of *renter size* on *supply quantity* also differ significantly between nonprofessional hosts and professional hosts (Difference = 0.0166, $p < 0.000$). Fig. 2(c) shows that with a shock of renter size, the quantity of nonprofessional hosts (vs. professional hosts) increases more, indicating that adding a number of renters would attract more nonprofessional

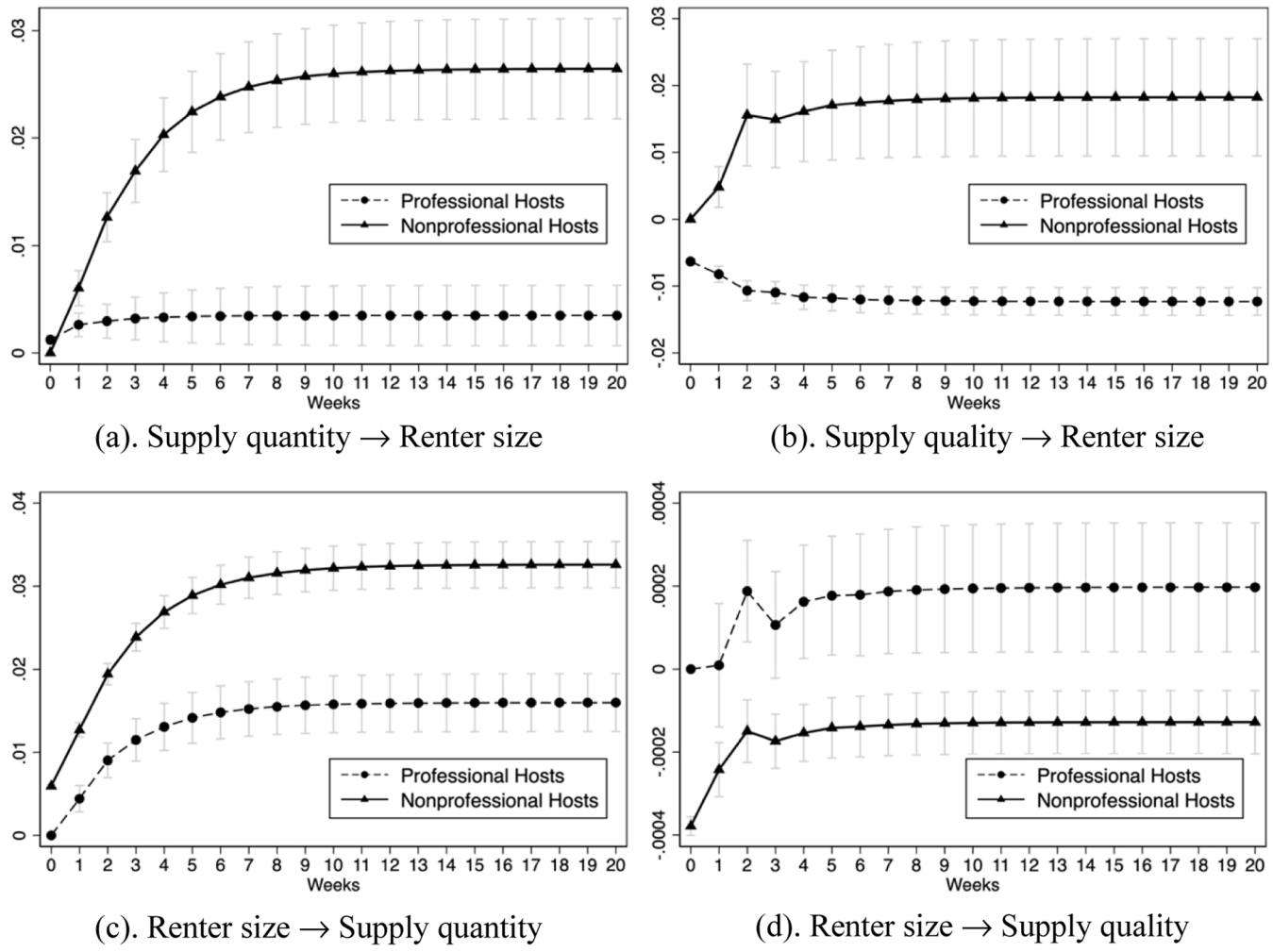


Fig. 2. Dynamics between the host side (supply quantity and quality) and the renter side
Notes: Connected lines represent the mean cumulative orthogonalized IRFs, along with the 95 % confidence interval.

hosts (vs. professional hosts). That is, in the quantity effects of CNEs, the renter-to-nonprofessional CNEs are stronger than the renter-to-professional CNEs. Hence, H1b is supported.

5.2.2. Heterogeneous CNEs between supply quality and renter size

Regarding the host-to-renter CNEs, the results in Table 6 and Figure 2 (b) show that the effects of supply quality on renter size significantly differ depending on professional and nonprofessional hosts. Specifically, *nonprofessional host quality* has a larger impact on *renter size* than *professional host quality* does (Difference = 0.0306, $p < 0.000$), which suggests that the quality enhancement of nonprofessional hosts (vs. professional hosts) could attract more renters. In other words, in the quality effects of CNEs, the nonprofessional-to-renter CNEs are stronger than the professional-to-renter CNEs. Hence, H2a is supported.

In addition, we find some counterintuitive but reasonable effects. That is, the quality enhancement of professional hosts has a significantly negative effect on renter size. This is plausible when professional hosts have opportunistic behaviors like disintermediation [92] or platform exploitation [93]. For example, to maximize profits, high-quality professional hosts, who can easily build trust with renters, are likely to tempt renters to leave the platform and trade offline. Similarly, some professional hosts may exploit platforms as symbolic touchpoints, showcasing high-quality properties to attract and migrate online renters toward their own offline channels. These potential opportunistic behaviors are recognized as increasing the likelihood of renter churn, thereby impeding renter growth.

Regarding renter-to-host CNEs, Table 6 and Fig. 2(d) suggest that the effects of renter size on supply quality are heterogeneous for professional and nonprofessional hosts. Specifically, the shock of increased renter size has significantly weaker effects on nonprofessional host quality than on professional host quality (Difference = −0.0004, $p < 0.000$). In other words, in the quality effects of CNEs, the renter-to-professional CNEs are stronger than the renter-to-nonprofessional CNEs. Thus, H2b is supported.

Additionally, we find the shock of increased renter size has significant negative effects on the quality of nonprofessional hosts. There are two potential reasons for such effects. First, the growth of renters may stimulate the entry of low-quality nonprofessional hosts [45], which lowers the overall quality of nonprofessional suppliers. Second, the growth of renters means that hosts need to invest more time and effort in interacting with renters. As such, it has become increasingly costly for nonprofessional hosts with limited resources to maintain sufficient quality to meet renters' expectations, which results in a decline in the quality of nonprofessional hosts.

5.3. Robustness checks

We further conduct several robustness checks and find consistent results. First, based on Abrigo and Love [86], we need to conduct the eigenvalue test to check the stability conditions of IRF estimates. Covariate estimates are stable only if all the eigenvalues are smaller than 1 (i.e., Modulus < 1). According to the results in Table B1, the IRF

estimates are stable and consistent. Thus, our hypothesis testing results based on estimated IRFs are suggested to be stable [85].

Second, following recent empirical studies (e.g., [83,82]), we choose a different number of lags as instruments to reestimate the PVAR model and rebuild the IRFs to check the robustness of the results. We use 10 lags as instruments in the main results, as we obtain the optimal lags under this setting. Notably, it seems arbitrary to select the number of lags as instruments in the PVAR model [82]. Extant literature suggests it's necessary to show that the main results are not sensitive to the specifications of instruments (e.g., [83]). Therefore, following Li et al. [83], we rerun the PVAR model with twenty lags as instruments. The results shown in Table B2 are consistent with our hypothesis testing results.

Third, we use the number of properties as an alternative measure that characterizes the supply quantity, as some studies suggest using the number of products to measure the supply quantity [30,19]. Ideally, we should remove duplicate listings of the same property to calculate the number of unique properties. For example, the same property may be assigned to multiple professional hosts within the same agency, leading to redundant and repeated listings. However, due to data limitations, we are unable to accurately identify truly repeated listings. Instead, we rely on identifying "similar" properties, which may include both repeated listings—where the same property is listed by multiple hosts—and distinct but highly similar properties, such as units located in the same residential community with comparable characteristics.

Specifically, we compare each listing in the listing dataset to identify whether identical property characteristics exist across different listings, excluding host information. These characteristics include the community location, layout, size, rent, room type (e.g., primary bedroom, second bedroom, entire unit), and rental type (e.g., shared, entire). Based on this comparison, we identify 216,188 properties with at least one similar listing. Then, we remove these properties from the dataset. Using the refined dataset, we calculate the number of nonprofessional hosts' properties in district i in week t, to measure $NonproProperty_{it}$. Also, we calculate the number of professional hosts' properties in district i in week t to measure $ProProperty_{it}$. We rerun the PVAR model shown in Eq. (1) with the measures $NonproQuantity_{it}$ and $ProQuantity_{it}$ replaced by $NonproProperty_{it}$ and $ProProperty_{it}$, respectively. The results, as summarized in Table B3, remain qualitatively consistent with our main findings.

5.4. Additional analyses

We conduct additional analyses to further develop insights that complement our main findings. As mentioned in Section 3, regarding supplier-to-consumer CNEs, a potential reason for the heterogeneous CNEs driven by different suppliers is that adding a number of nonprofessional suppliers (vs. professional suppliers) would help platforms increase supply diversity and decrease product prices. Regarding consumer-to-supplier CNEs, a potential reason for the heterogeneous CNEs among different suppliers is that adding a number of consumers would increase platform supply diversity, thereby providing nonprofessional suppliers (vs. professional suppliers) with more diverse opportunities. Therefore, to further verify these insights, we would examine the potential impacts of adding different suppliers and consumers.

First, using the district-level sample derived from the panel data mentioned in the PVAR analysis, we initially conducted exploratory data analyses by drawing scatter plots of related variables. Specifically, we create four related variables: $Nonprofessional_Ratio_j$, $\ln(RenterSize)_j$, $Supply\ diversity_j$, and $\ln(AvgRent)_j$. $Nonprofessional_Ratio_j$ measures the ratio of nonprofessional hosts in district j and a higher value means adding more nonprofessional hosts (vs. professional hosts) in district j. $\ln(RenterSize)_j$ is measured by the natural logarithm of renter size in district j. $Supply\ diversity_j$ is operationalized using a Herfindahl-like index measure. Similar to the calculation process of *Product scope* in the model-free analysis, we classify the properties into 162 unique categories. Then we define $HI_j = \sum_{i=1}^{N} \left(\frac{x_{ij}}{x_j}\right)^2$, where x_i is the number of properties in the ith category of properties in district j, x_j is the total number of properties in district j, $\frac{x_{ij}}{x_j}$ is the share of the ith category of properties. In this way, a lower level of HI_j suggests a higher degree of supply diversity in district j. To make the resulting interpretation more intuitive, we let $Supply\ diversity_j = 1\text{-}HI_j$ (as 1 is the HI_j maximum value). Therefore, a higher value of $Supply\ diversity_j$ means a higher degree of supply diversity in district j. $\ln(AvgRent)_j$ is measured by the natural logarithm of the average rent of properties in district j. A higher value of $\ln(AvgRent)_j$ indicates a higher price level on the market.

As shown in Fig. 3, we plot the relationships among the four variables. Figure 3(a) shows a positive relationship between $Nonprofessional_Ratio_j$ and $Supply\ diversity_j$, indicating that adding nonprofessional hosts (vs. professional hosts) would increase the supply diversity on the market. Figure 3(b) shows a negative relationship between $Nonprofessional_Ratio_j$ and $\ln(AvgRent)_j$, which indicates that adding nonprofessional hosts (vs. professional hosts) would decrease the average price on the market. Fig. 3(c) shows a positive relationship between $\ln(RenterSize)_j$ and $Supply\ diversity_j$, indicating that adding renters would increase the supply diversity on the market. Consistent with our theoretical development, these intuitive patterns demonstrate potential explanations for heterogeneous CNEs between different suppliers and consumers.

Second, we further conducted regression analyses using the district-week-level sample mentioned in the PVAR analysis to find more convincing evidence. Specifically, four key variables are created. $Nonprofessional_Ratio_{jt}$ measures the ratio of nonprofessional hosts in district j in week t. $RenterSize_{jt}$ measures the renter size in district j in week t. $Supply\ diversity_{jt}$ measures the supply diversity in district j in week t. $AvgRent_{jt}$ measures the average rent per property in district j in week t. Additionally, several control variables are also created to control district-level time-variant characteristics. $\#Property_{jt}$ measures the number of properties in district j in week t. $HostSize_{jt}$ measures the host size in district j in week t. $TimetoMatch_{jt}$ measures the average months needed to rent out a property in district j in week t. $\%Match_{jt}$ measures the ratio of rented-out properties in district j in week t. $\#Match_{jt}$ measures the number of rented-out properties in district j in week t. To address the skew in distributions, we use the natural logarithm of $RenterSize_{jt}$, $AvgRent_{jt}$, $\#Property_{jt}$, and $HostSize_{jt}$ values.

As summarized in Table C1 in Appendix C, the results show that $Nonprofessional_Ratio_{jt}$ has a significant positive impact on $Supply\ diversity_{jt}$ ($\beta = 0.094$, $p < 0.001$), and a significant negative impact on $\ln_AvgRent_{jt}$ ($\beta = -2.385$, $p < 0.001$). Besides, column (1) in Table C1 shows a significant positive impact of $\ln_RenterSize_{jt}$ on $Supply\ diversity_{jt}$ ($\beta = 0.012$, $p < 0.001$). Consistent with our arguments in H1 and H2, these findings are suggestive evidence helping to understand why CNEs between different suppliers and consumers may be heterogeneous. That is, regarding supplier-to-consumer CNEs, adding nonprofessional suppliers (vs. professional suppliers) could better help platforms drive consumer growth by increasing supply diversity and decreasing product prices on the platforms. Regarding consumer-to-supplier CNEs, adding consumers would increase supply diversity on the platforms due to increased heterogeneous needs, thereby resulting in a larger attractiveness to nonprofessional suppliers (vs. professional suppliers). Furthermore, the suggestive evidence also sheds light on the potential superiority of nonprofessional suppliers in platform growth. While professional suppliers generally have higher quality and perform better (e.g., more revenue) than nonprofessional suppliers, it is important to acknowledge that nonprofessional suppliers contribute significantly to consumer growth by providing unique offerings and affordable prices.

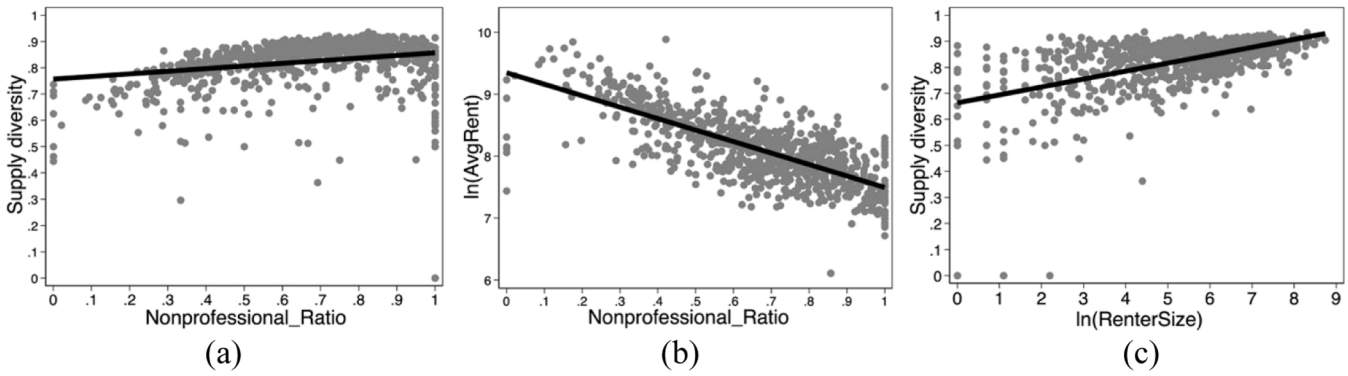


Fig. 3. Scatter plots of key attributes at the district level
Notes: Gray dots represent the value of attributes located on the vertical axis. Solid lines represent the fitted value between attributes located on the horizontal and vertical axis.

6. Discussion

6.1. Conclusion

Understanding CNEs is crucial given their strategic roles in platform growth and governance [9,11,13,19]. However, existing literature offers limited insights into whether and how CNEs between suppliers and consumers differ across different suppliers, which is critical in light of recent practical and academic concerns on nonprofessional and professional suppliers on P2P platforms. This study surfaces the potential heterogeneity of CNEs between such two types of suppliers and consumers. Using the unique data collected from a platform matching hosts and renters, we conduct the PVAR analysis and obtain the main findings summarized in Table 7. Specifically, regarding supplier-to-consumer CNEs, our results reveal that both the quantity and quality of nonprofessional suppliers (vs. professional suppliers) have stronger CNEs on consumer size. Regarding the consumer-to-supplier CNEs, the consumer size has stronger CNEs on the quantity of nonprofessional suppliers (vs. professional suppliers) but has weaker CNEs on the quality of nonprofessional suppliers (vs. professional suppliers).

These findings can be generalized to P2P platforms with high-level supplier heterogeneity, although the empirical context in this study is a platform in real estate rental properties. Generally, the practice of allowing both nonprofessional and professional suppliers on board is widespread for P2P platforms. On such platforms, high-level supplier heterogeneity would lead to differentiated consumer benefits, thereby resulting in heterogeneous CNEs. For example, on platforms like eBay and Taobao, firm sellers (i.e., professional sellers), who have richer resources than individual sellers (i.e., nonprofessional sellers), tend to offer general and popular products, while individual sellers mainly offer niche products and services. Therefore, akin to our findings, the responses of the seller side and the buyer side to each other may differ in seller type. Similarly, in freelance markets like Fiverr, amateur freelancers are distinct from professional freelancers, which may result in heterogeneous CNEs. Furthermore, these platforms also exhibit the pattern that adding nonprofessional suppliers could help meet heterogeneous needs and facilitate rapid platform growth [20].

Table 7
Hypothesis testing results and main findings.

Supplier type	Supplier-to-Consumer CNEs		Consumer-to-Supplier CNEs	
	Supply quantity (H1a)	Supply quality (H2a)	Supply quantity (H1b)	Supply quality (H2b)
Nonprofessional	Stronger	Stronger	Stronger	Weaker
Professional	Weaker	Weaker	Weaker	Stronger
Heterogeneous	YES	YES	YES	YES

6.2. Theoretical implications

Our empirical findings yield three important theoretical implications for P2P platforms with high-level supplier heterogeneity. First, to the best of our knowledge, ours is the first empirical study to examine the heterogeneous CNEs driven by different suppliers, which contributes to CNEs literature in two-sided platforms. Though some recent studies have investigated heterogeneous CNEs, most of them assume that all suppliers are homogeneous [20,18,19,41]. Our findings deepen the theoretical understanding of CNEs and essentially echo the call in recent studies for the exploration of supplier heterogeneity in CNEs (e.g., [11]). Furthermore, this study enriches platform governance literature on how to better stimulate CNEs [36,34,10,38]. Indeed, given our results of heterogeneous CNEs between two types of suppliers and consumers, platforms could better stimulate CNEs by tailoring governance strategies to different suppliers.

Second, the insights from our study also contribute to the theoretical implications of nonprofessional and professional suppliers. Extant literature has investigated the impacts of the two types of suppliers on platform quality and performance, primarily highlighting the superiority of professional suppliers [45,1,2,46,26]. Limited insights are offered into the role of nonprofessional suppliers. This study contributes by exploring their impacts on platform growth and unveiling the potential superiority of nonprofessional suppliers. Specifically, our results suggest that adding nonprofessional suppliers (vs. professional suppliers) could help platforms attract more consumers by increasing supply diversity and decreasing product prices on platforms. These findings align with practical observations and help explain the growth of certain platforms. For example, the growth of Airbnb's accommodation business has primarily been driven by nonprofessional hosts worldwide [4]. Furthermore, our nuanced insight implies that the roles of the two types of suppliers are contingent on the specific platform objectives, offering valuable implications for optimizing platform strategies.

Third, this study expands the theoretical understanding of two critical supply-side attributes—quantity and quality—in capturing CNEs. While prior literature has extensively examined quantity in CNEs, quality is regarded as another critical but understudied attribute on the supply side, which would influence the consumption benefits of consumers, thereby influencing the strength of CNEs [5,52,53]. Besides, extant literature calls for more nuanced conceptualizations of CNEs by unveiling other attributes beyond the sheer quantity. Therefore, this study contributes by using both quantity and quality attributes on the supply side to help capture CNEs, which provides a more nuanced perspective on how heterogeneous CNEs emerge. Furthermore, our findings reveal that changes in the quantity and quality of nonprofessional and professional suppliers affect consumer benefits in distinct ways, leading to varied strengths of CNEs. This differentiation highlights the need for platform managers to tailor governance strategies to

account for both supplier type and supply attribute.

6.3. Practical implications

The characterization of heterogeneous CNEs driven by different suppliers generates three implications for P2P platforms with high-level supplier heterogeneity. First, to stimulate strong CNEs, platforms usually need to decide which groups to prioritize in resource commitment [19]. Overlooking the potential supplier heterogeneity may result in a waste of resources and low effectiveness of governance strategies. Given our results that both the quantity and quality of nonprofessional suppliers (vs. professional suppliers) can stimulate stronger CNEs on consumers, managers aiming to grow the platforms are suggested to invest relatively more resources in attracting nonprofessional suppliers (vs. professional suppliers) and supporting their quality enhancements. For example, Fiverr uses "Fiverr learn" to help amateur freelancers improve their skills and service quality. In this way, platforms could not only leverage the positive feedback loops between consumer growth and supplier growth, but also mitigate the feedback reductions between consumer growth and quality enhancement of nonprofessional suppliers (as adding consumers stimulates weaker CNEs on quality enhancement of nonprofessional suppliers than vice versa).

Second, platforms should be aware that the strategic management of nonprofessional and professional suppliers is contingent on the specific platform objective. Although prior literature often emphasizes the greater roles of professional suppliers (vs. nonprofessional suppliers) in platform quality and performance, our results underscore the greater importance of nonprofessional suppliers in driving platform growth. This is plausible as platform growth doesn't imply platform performance and vice versa [94]. Therefore, when facing different objectives, platform managers should seriously consider the relative roles of different suppliers and make proper strategic decisions. Doing so can enhance the platform's ability to manage different suppliers and help achieve strategic goals more effectively. For example, when aiming to stimulate platform growth, platforms could attract more nonprofessional suppliers to better grow the user base.

Third, a key practical implication is the need to develop governance capabilities that manage both the quantity and quality of nonprofessional and professional suppliers. While prior CNEs literature has emphasized the importance of supply quantity and quality [5,52,62,53,40], limited research has explored the simultaneous consideration of these two attributes along with supplier heterogeneity. Our empirical results highlight the need for managing both the quantity and quality of nonprofessional and professional suppliers to leverage CNEs and drive platform growth. This dual-focus perspective (i.e., considering both supplier type and supply attribute) offers actionable insights for managing heterogeneous suppliers more strategically.

6.4. Limitations and future research

This study has some limitations and creates opportunities for future research. First, our findings mainly reveal the nature of heterogeneous CNEs on P2P platforms with high-level supplier heterogeneity. As for P2P platforms with low-level supplier heterogeneity, suppliers' differences may be too minimal to produce distinct impacts on consumer benefits, thereby leaving the nature of CNEs driven by different suppliers unclear. For example, on Uber, taxi drivers (i.e., professional drivers) and private drivers (i.e., nonprofessional drivers) have similar skills and provide similar ride services [95]. Moreover, on non-P2P platforms where the classifications of suppliers vary and differ from this study, the nature of CNEs driven by different suppliers remains unknown. For example, Zipcar acts as a first-party supplier offering car-rental service directly to consumers, while Amazon functions both as a first-party seller and as intermediary platform to facilitate third-party sellers (including nonprofessional and professional sellers). Future research could extend the insights on heterogeneous CNEs by exploring these research contexts.

Second, while extant literature suggests that professional suppliers (vs. nonprofessional suppliers) may play a more significant role in platform performance, our study reveals the superiority of nonprofessional suppliers in platform growth. Such mixed findings regarding the roles of the two types of suppliers are plausible as platform growth and performance are distinct and not necessarily correlated [94]. This implies that different suppliers potentially have varying impacts on consumer acquisition and purchase. Future research can deepen the insights by simultaneously considering the multiple platform objectives and empirically examining the underlying impacts of the two types of suppliers on various consumer behaviors. Furthermore, this study primarily focuses on the role of supplier type rather than the underlying attributes and differences of suppliers. Future studies could contribute by empirically examining how two types of suppliers differ in various attributes and how these differences matter.

Third, the role of platform access control to different suppliers, an important governance mechanism [9,96], is not addressed in this study due to data limitations. Although we have demonstrated the superiority of nonprofessional suppliers, our results do not suggest that only retaining nonprofessional suppliers is optimal for platform growth. While it is common for P2P platforms to include both types of suppliers (e.g., Airbnb, Fiverr, eBay), some platforms exclusively grant access to one group of suppliers. For example, Houzz, an online platform focusing on home design, only allows professional designers to offer services. Examining the impacts of access control on the two types of suppliers could provide valuable insights into supplier management. Future research could thus focus on platform governance strategies and specifically consider the implications of different suppliers. Furthermore, due to data limitations, this study primarily uses consumer complaint data to capture the quality of suppliers. While this measure offers valuable insights into one aspect of quality—consumer dissatisfaction—it does not fully account for other dimensions such as product or service value. Therefore, future research could develop more comprehensive quality metrics to accurately capture supplier quality.

CRediT authorship contribution statement

Wenyu Yang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peijian Song:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Conceptualization. **Yinglin Ruan:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

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Appendix A. Screenshots of Property Page

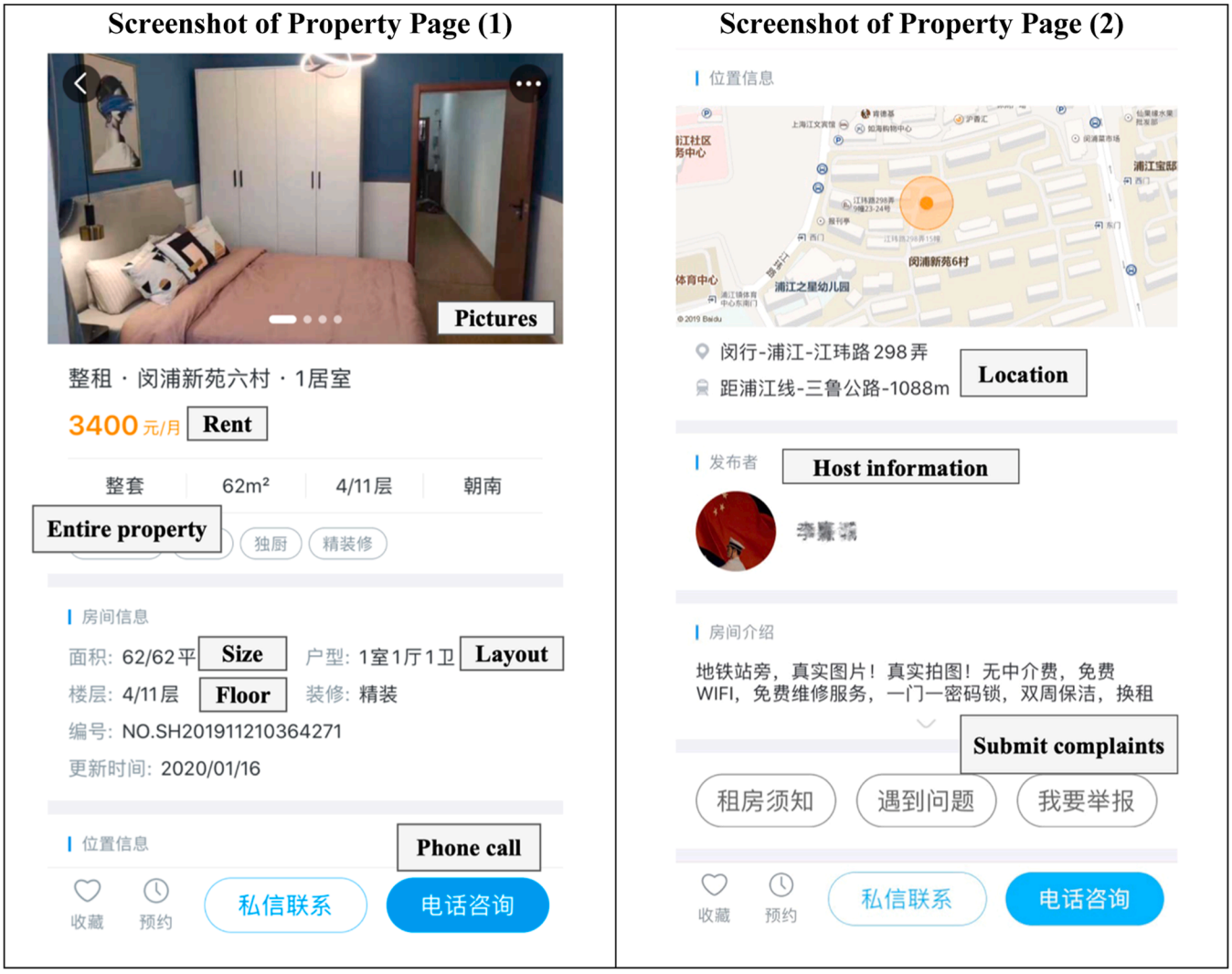


Fig. A1. Screenshots of Property Page.

Fig. A1 provides two screenshots of a property on the focal platform from the renters' perspective. Prospective renters could browse detailed property information such as the rent, rent type (i.e., entire property or shared property), size, floor, layout, and location. Then, prospective renters could call the host to learn more about the property's information. On the page of each listed property, there is a link for renters to submit a complaint as shown in the screenshot of property page (2) in Fig. A1. In this way, renters could complain about any reason for their dissatisfaction with the property or the host's service.

Appendix B. Robustness Checks

Table B1
Eigenvalue stationarity tests.

	Real	Imaginary	Modulus
$NonproQuantity_{i,t-1}$	0.615	0.000	0.615
$NonproQuantity_{i,t-2}$	0.356	0.000	0.356
$ProQuantity_{i,t-1}$	−0.291	0.000	0.291
$ProQuantity_{i,t-2}$	−0.198	0.000	0.198
$NonproQuality_{i,t-1}$	0.152	0.000	0.152
$NonproQuality_{i,t-2}$	0.081	0.000	0.081
$ProQuality_{i,t-1}$	−0.118	0.324	0.345
$ProQuality_{i,t-2}$	−0.118	−0.324	0.345
$RenterSize_{i,t-1}$	−0.227	−0.220	0.316
$RenterSize_{i,t-2}$	−0.227	0.220	0.316

Table B2
Using 20 lags as instruments.

Panel A: Host-to-Renter CNEs			
	Host Type		Difference
Paths	Nonprofessional	Professional	
Supply quantity → Renter size (H1a)	0.0316**	0.0058*	0.0258***
	(0.003)	(0.002)	(0.003)
Supply quality → Renter size (H2a)	0.0053***	−0.0102***	0.0155***
	(0.003)	(0.001)	(0.003)
Panel B: Renter-to-Host CNEs			
	Host Type		Difference
Paths	Nonprofessional	Professional	
Renter size → Supply quantity (H1b)	0.0363***	0.0165***	0.0197***
	(0.001)	(0.002)	(0.002)
Renter size → Supply quality (H2b)	−0.0001**	0.0001*	−0.0003***
	(0.000)	(0.000)	(0.000)

Notes: The table presents cumulative orthogonalized IRFs at week 20. Robust standard errors are presented in parentheses. Significance levels: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Table B3
Using number of properties as alternative measurement of supply quantity.

Panel A: Host-to-Renter CNEs			
	Host Type		Difference
Paths	Nonprofessional	Professional	
Supply quantity → Renter size (H1a)	0.0262**	0.0084*	0.0178***
	(0.002)	(0.002)	(0.003)
Supply quality → Renter size (H2a)	0.0185***	−0.0159***	0.0344***
	(0.004)	(0.001)	(0.004)
Panel B: Renter-to-Host CNEs			
	Host Type		Difference
Paths	Nonprofessional	Professional	
Renter size → Supply quantity (H1b)	0.0328***	0.0246***	0.0082***
	(0.001)	(0.002)	(0.002)
Renter size → Supply quality (H2b)	−0.0001*	0.0001*	−0.0003**
	(0.000)	(0.000)	(0.000)

Notes: The table presents cumulative orthogonalized IRFs at week 20. Robust standard errors are presented in parentheses. Significance levels: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Appendix C. Additional Analyses

Table C1
Potential impacts of adding renters and nonprofessional hosts.

		(1)	(2)
		DV: Supply diversity	DV: ln_AvgRent
Independent variables			
	Nonprofessional_Ratio	0.094***	−2.385***
		(0.007)	(0.282)
	ln_RenterSize	0.012***	−0.062
		(0.001)	(0.040)
Control variables			
	ln_#Property	0.021***	−0.429***
		(0.005)	(0.068)
	ln_HostSize	−0.049***	−0.466***
		(0.005)	(0.081)
	TimetoMatch	0.003***	−0.091***
		(0.001)	(0.014)
	%Match	−0.147***	4.361***
		(0.019)	(0.645)
	#Match	0.000**	0.004***
		(0.000)	(0.000)
	Constant	0.627***	15.143***
		(0.036)	(0.838)
	Time FE	Included	Included
	District FE	Included	Included
N		103,396	103,396
R^2		0.401	0.269

(*continued on next page*)

Table C1 (*continued*)

	(1)	(2)
	DV: Supply diversity	DV: ln_AvgRent
Adjust R^2	0.400	0.268
AIC	−379,771.902	410,209.032
BIC	−378,082.203	411,898.731
Log-likelihood	190,062.951	−204,927.516

Notes: Robust standard errors are presented in parentheses. Significance levels: $^{*}p < 0.01$; $^{**}p < 0.05$; $^{***}p < 0.001$.

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